NXT Studio

IOT Data Analytics Internship

Week 7 Report

Submitted By

Name: Saniya Syed

Course: B.Tech CSE (INTERNET OF THINGS)

Collage: Jain (Deemed-to-be University)

Submitted To

NXT Studio

Internship Duration

12 Weeks

Report

Week 7 Anomaly Detection in IoT Data

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Objective

The objective of this experiment is to detect unusual sensor behavior in an IoT dataset using different anomaly detection techniques. Three methods—Threshold-Based Detection, Isolation Forest, and Local Outlier Factor (LOF)—are applied to identify abnormal sensor readings and compare their performance.

Dataset Description

The dataset used in this experiment is an IoT sensor dataset containing **2311 sensor records**. It includes the following columns:

- Date
- Temperature
- Humidity
- Verdict

The dataset records environmental sensor readings over time and is used to identify abnormal or unusual observations using anomaly detection algorithms.

Threshold-Based Detection

Threshold-Based Detection is a simple anomaly detection technique where predefined limits are used to identify abnormal values. In this experiment, temperature values below **15°C** or above **40°C** are considered anomalies.

Explanation:
The Threshold-Based Detection method identified **2105 normal** sensor readings and **206 anomalous** readings based on predefined temperature limits.

```python
df["Threshold_Anomaly"] = (
    (df["temperature"] > 40) |
    (df["temperature"] < 15)
)

df["Threshold_Anomaly"].value_counts()
```

	count
Threshold_Anomaly	
False	2105
True	206

dtype: int64

Figure 1: Threshold-Based Anomaly Detection Results

Explanation:
The scatter plot displays normal sensor readings in **blue** and anomalous readings in **red**, making it easy to identify unusual temperature values over time.

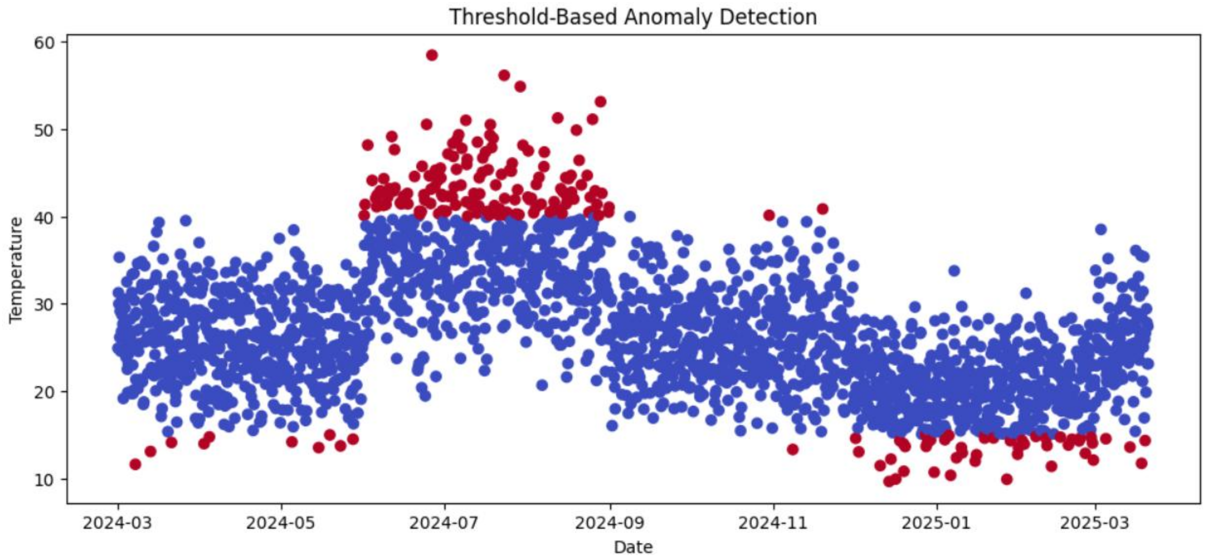
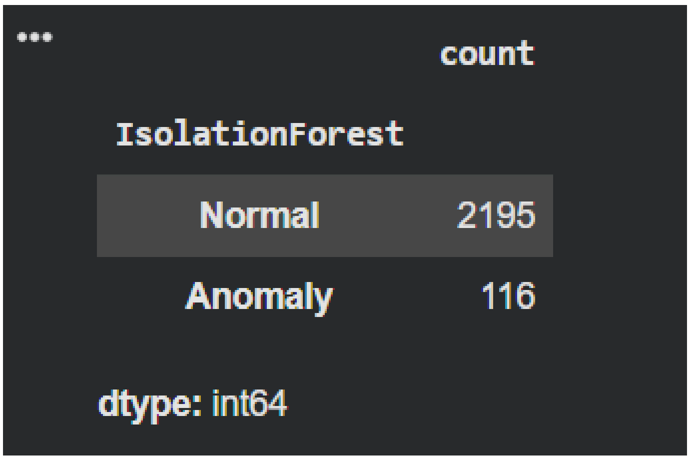


Figure 2: Threshold-Based Anomaly Detection Visualization.

Isolation Forest

Isolation Forest is an unsupervised machine learning algorithm that detects anomalies by isolating unusual observations. It identifies data points that differ significantly from the majority of the dataset.



	count
IsolationForest	
Normal	2195
Anomaly	116

dtype: int64

Figure 3: Isolation Forest Detection Results

Explanation:
The Isolation Forest algorithm identified **2195 normal** sensor readings and **116 anomalous** readings by analyzing the overall data distribution.

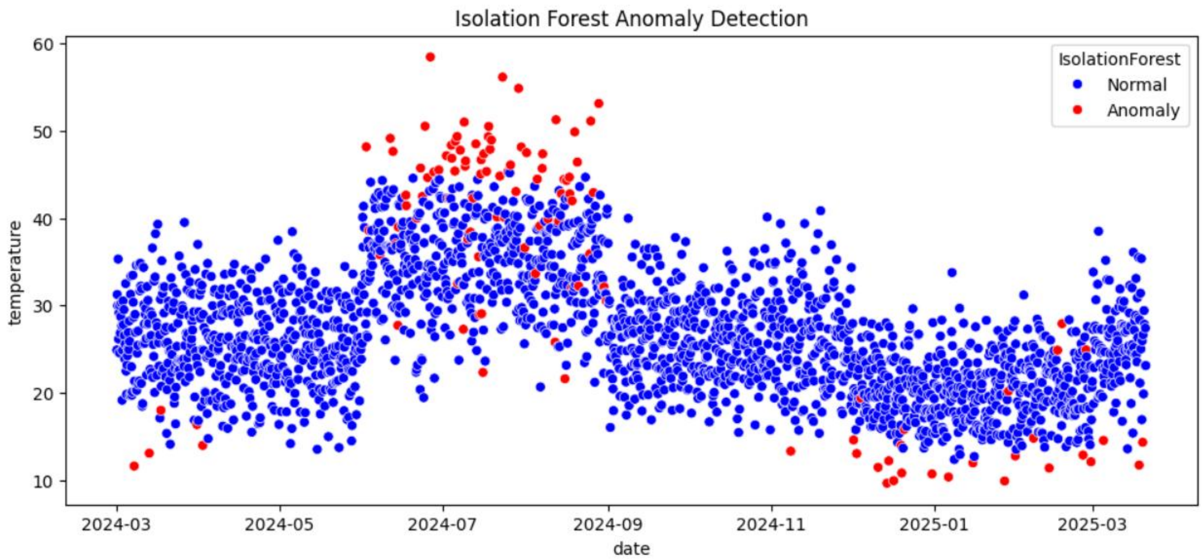


Figure 4: Isolation Forest Anomaly Detection Visualization

Explanation:
The scatter plot highlights normal sensor readings in **blue** and detected anomalies in **red**, showing the abnormal temperature values identified by the Isolation Forest algorithm.

Local Outlier Factor (LOF)

Local Outlier Factor (LOF) is an unsupervised anomaly detection algorithm that identifies abnormal data points by comparing the local density of each observation with that of its neighboring points. Points with significantly lower density are classified as anomalies.

LOF	count
Normal	2195
Anomaly	116

dtype: int64

Figure 5: Local Outlier Factor (LOF) Detection Results

Explanation:

The LOF algorithm identified **2195 normal** sensor readings and **116 anomalous** readings by analyzing the local density of the dataset.

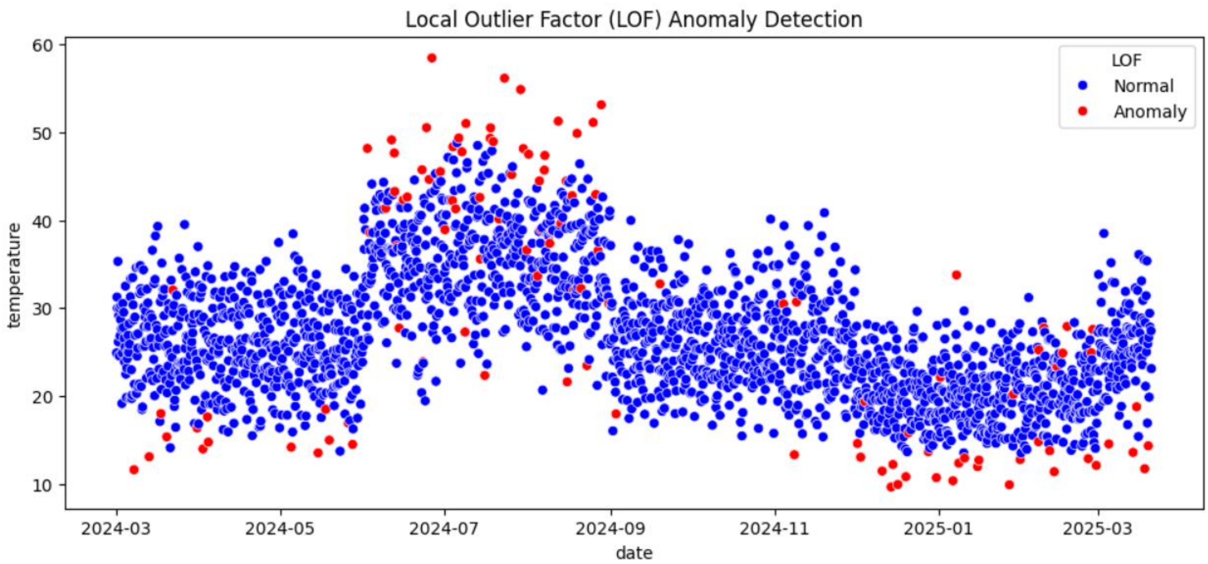


Figure 6: Local Outlier Factor (LOF) Anomaly Detection Visualization

Explanation:

The scatter plot displays the anomaly detection results using the LOF algorithm. **Blue points** represent normal sensor readings, while **red points** indicate anomalous readings detected based on differences in local data density.

Comparison of Anomaly Detection Methods

Method	Normal	Anomaly
Threshold-Based Detection	2105	206
Isolation Forest	2195	116
Local Outlier Factor (LOF)	2195	116

Observations

- Threshold-Based Detection identified anomalies using predefined temperature limits (>40°C or <15°C).
- Isolation Forest detected anomalies by identifying unusual patterns in the dataset.
- Local Outlier Factor (LOF) detected anomalies based on the local density of neighboring data points.
- Isolation Forest and LOF detected fewer anomalies than the Threshold-Based method, indicating more selective and pattern-based detection.

Conclusion

This experiment successfully applied three anomaly detection techniques to an IoT sensor dataset. Threshold-Based Detection used fixed temperature limits, while Isolation Forest and Local Outlier Factor (LOF) used machine learning algorithms to identify unusual sensor behavior. The analysis demonstrated that machine learning methods provide more intelligent and reliable anomaly detection, making them suitable for real-world IoT monitoring systems.